

Tide retrospective: *2D numerator asymptotic for the quartic-sextic potential (Z1)*

Daniel Murfet

18 May 2026

Abstract

The cross-project tide-candidate survey labelled the V2 follow-up “numerator-side asymptotic”. This Tide discharges it: the missing literal interpretation of the original survey wording, about the unnormalised 2D integral $\iint x^{2j}y^{2k} e^{-tL} dx dy$ scaling as $t^{-(2j+1)/4-(2k+1)/6}$. The packaging mirrors V2: an exact reformulation, a rescaled limit, and an asymptotic- equivalence form.¹ The proof is one factorisation through separability plus two 1D moment closed forms plus the same `rpow` algebra V2 used. 157 lines, single-attempt clean build, with zero `sorry`, axiom, or `native_decide`. Mirror- shaped tide; GPT consult skipped per the tide skill’s preflight, as the candidate is structurally identical to V2 and the closed form was verified numerically at ten test points to relative error $\sim 10^{-16}$.

1 Setting

Z1 is the literal completion of the V2 story: V2 packaged T1’s exact moment closed form into asymptotic form; Z1 does the same for the *unnormalised numerator integral*. The two objects differ by the 2D partition function $\iint e^{-tL} dx dy$, which itself scales as $t^{-1/4-1/6} = t^{-5/12}$; the moment scaling $t^{-(j/2+k/3)}$ and the numerator scaling $t^{-((2j+1)/4+(2k+1)/6)}$ differ by exactly $t^{-5/12}$, as expected.

Two retrospective-time observations made Z1 cheaper than the survey estimated:

- The 2D-numerator factor lemma already exists in the seabed. The survey direction guessed “likely needs to be proved alongside, by direct factorisation of the 2D integral”. In fact, for any `addSeparable` potential $U + V$ and any separable observable, the 2D integral factors as $\iint f(x)g(y)e^{-t(U+V)} = (\int f e^{-tU})(\int g e^{-tV})$.²
- The 1D exact moments are also already in the seabed: `Laplace.OneD.quartic_moment_even` and `Laplace.OneD.sextic_moment_even`. T1 used them; Z1 uses them again, this time directly rather than via the `gibbsExpectation` layer.

This Tide is therefore the cleanest possible mirror of V2: the same three-theorem packaging with different building blocks for the exact reformulation step.

¹The latter two via `Tendsto` and `Asymptotics.IsEquivalent` respectively.

²In `Laplace/TwoD/AddSeparable.lean` as `integral_separable_addSeparable`.

2 The thing formalised

Three theorems in `Laplace.TwoD`:

1. **Exact reformulation.** For $t > 0$ and $j, k \in \mathbb{N}$,

$$\int \int x^{2j} y^{2k} e^{-tL(x,y)} dx dy = C'(j, k) \cdot t^{-((2j+1)/4 + (2k+1)/6)}$$

with

$$C'(j, k) := \frac{1}{6} \cdot 24^{(2j+1)/4} \cdot 720^{(2k+1)/6} \cdot \Gamma((2j+1)/4) \cdot \Gamma((2k+1)/6).$$

2. **Rescaled Tendsto.** As $t \rightarrow \infty$,

$$t^{(2j+1)/4 + (2k+1)/6} \cdot \int \int x^{2j} y^{2k} e^{-tL} \rightarrow C'(j, k).$$

3. **IsEquivalent.** At `atTop`,

$$\int \int x^{2j} y^{2k} e^{-tL} \sim C'(j, k) \cdot t^{-((2j+1)/4 + (2k+1)/6)}.$$

Named constant: `quarticSexticNumeratorConst`. File: sibling to V2's moment-asymptotic file, in the same `Laplace/TwoD/` folder.³ Added to project root.

3 Proof strategy

The exact reformulation pipeline:

1. Apply `integral_separable_addSeparable` with $f = x \mapsto x^{2j}$, $g = y \mapsto y^{2k}$, integrability witnesses `quartic_integrable_pow_pot` and `sextic_integrable_pow_pot`. This factors the 2D integral into a product of two 1D integrals, but with the potentials still in `quarticPotential` / `sexticPotential` form.
2. Rewrite each 1D integrand from `quarticPotential` `x` to the explicit `x^4 / 24` form (and similarly for sextic), via the `_apply @[simp]` lemmas. Each is a small `congr 1; ext; ...; ring` block; mirror of T1's template.
3. Apply `quartic_moment_even` and `sextic_moment_even` to replace each 1D integral with its $(\text{const} \times (24/t)^{\text{exp}} \times \Gamma)$ form.
4. Unfold `quarticSexticNumeratorConst`, then standard `Real.rpow` algebra: `Real.div_rpow` splits $(24/t)^a$ into $24^a/t^a$, `Real.rpow_add` (valid for $t > 0$) combines $t^{-a} \cdot t^{-b} = t^{-(a+b)}$, and `Real.rpow_neg` bridges negative exponents. `ring` closes.

The two asymptotic forms (`Tendsto` and `IsEquivalent`) follow from the exact reformulation by the same eventual-equality idiom V2 used. Both reduce to “the rescaled function is eventually equal to a constant” on the positive reals.

³Specifically `Laplace/TwoD/QuarticSexticNumeratorAsymptotic.lean`, alongside `QuarticSexticMomentAsymptotic.lean` from V2.

4 Roadblocks and resolutions

No roadblocks. The proof compiled green on first try. The lessons from V2 (no `field_simp` after explicit `rpow_add` normalisation; use `ring` at the bottom; `tendsto_congr'` + eventual equality for the asymptotic packagings) carried directly.

The structural insight made before writing any code — that the seabed already provides the 2D-numerator factor lemma — saved roughly a third of the expected work (the original Z1 estimate budgeted for proving a numerator factor lemma alongside, ~30 lines).

5 What was Mathlib, what was new

From Mathlib. The standard `Real.rpow` algebra suite,⁴ the `Tendsto-to-constant` idiom,⁵ and the `Asymptotics.IsEquivalent.refl` + congruence pair.⁶ Same Mathlib API V2 used.

From the seabed. Five seabed lemmas, all proved on prior tides: the 2D-numerator factor theorem from A2's `AddSeparable.lean`, two 1D integrability witnesses, two 1D `_moment_even` closed forms, and the two `_Potential_apply @[simp]` unfolds.⁷ Nothing else from the laplace seabed was load-bearing.

New here. Three theorems plus the named constant `quarticSexticNumeratorConst`. Architecturally identical to V2; the value is in completing the numerator-vs-moment parity.

6 Lessons

Mirror-shaped tides are first-class. A tide whose candidate is structurally identical to a just-landed predecessor *should* skip the GPT consult — the deliberation has already happened, and re-running it adds token cost without changing the output. The tide skill's preflight already names this path (“proceed-without-GPT”); Z1 is a clean exemplar. The discipline that justifies the skip is (a) a numerical sanity check covering at least 5–10 test points, and (b) explicit acknowledgement in the tide log that the candidate is mirror-shaped against the named predecessor.

Seabed dig before deliberation pays off. A 5-minute `grep` of `AddSeparable.lean` surfaced the existing 2D-numerator factor lemma, which the survey direction had guessed needed to be proved alongside Z1. Knowing this before drafting candidates cut the Tide cost estimate by a third and made the “mirror of V2” framing exact.

Moment vs numerator scaling differ by the partition. V2's moment scales as $t^{-(j/2+k/3)}$; Z1's numerator scales as $t^{-((2j+1)/4+(2k+1)/6)}$. The difference is exactly $t^{-5/12} = t^{-1/4-1/6}$, which is the partition-function scaling (quartic partition $\sim t^{-1/4}$, sextic $\sim t^{-1/6}$). Worth recording as a sanity-check identity for future asymptotic packagings.

⁴`Real.div_rpow`, `Real.rpow_add`, `Real.rpow_neg`, plus `ring`.

⁵`tendsto_const_nhds` and `Tendsto.congr'`.

⁶`Asymptotics.IsEquivalent.refl` and `IsEquivalent.congr_left`.

⁷Specifically `integral.separable.addSeparable`, `quartic.integrable.pow_pot`, `sextic.integrable.pow_pot`, `quartic.moment_even`, `sextic.moment_even`, `quarticPotential.apply`, and `sexticPotential.apply`.

7 Follow-ups

- *2D partition function asymptotic.* The natural completing third object: the 2D partition function $\iint e^{-tL}$ scales as $C'' \cdot t^{-1/4-1/6}$. Same three-theorem packaging, even smaller proof (no polynomial weights, just the partition product). ~50 lines.
- *Odd-numerator analogues.* For any odd index, the 2D numerator with x^{2j+1} or y^{2k+1} weight is identically zero by V1. The corresponding “asymptotic packaging” is the trivial $0 \sim 0$; skip.
- *Generic k -th-power potential 1D and 2D analogues.* The same template (1D moment closed form + 2D separability) works for $x^{2k}/(2k)!$ pure-monomial potentials in general. The 1D generic- k asymptotic is exactly N1; extending to 2D follows the Z1 template. Out of scope for this tide.
- *Promote `quarticSexticNumeratorConst` to `lean-common`.* Only with a third consumer; currently the named constant has one use site.

Acknowledgements

No GPT consult was used for Z1 (mirror-shaped tide off V2 per the tide skill’s preflight). The deliberation work was inherited from V2’s GPT consult, which set the three-theorem packaging shape and the `tendsto_congr`’ idiom; both apply unchanged here. The numerical sanity check at ten (j, k, t) values confirmed the $C'(j, k)$ formula to relative error $\sim 10^{-16}$ before the proof was written, and is preserved in the primer’s tide-log GPT-responses folder alongside this tide’s markdown log.